



DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

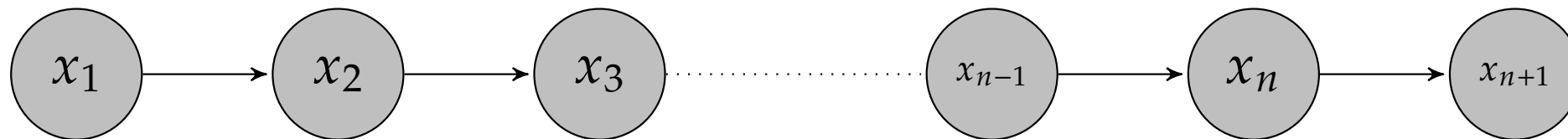
Probabilistic Machine Learning

Hidden Markov Models

Luca Bortolussi, University of Trieste

Sequential data and Markov Models

time series: financial, speech, weather, epidemiological data, ...

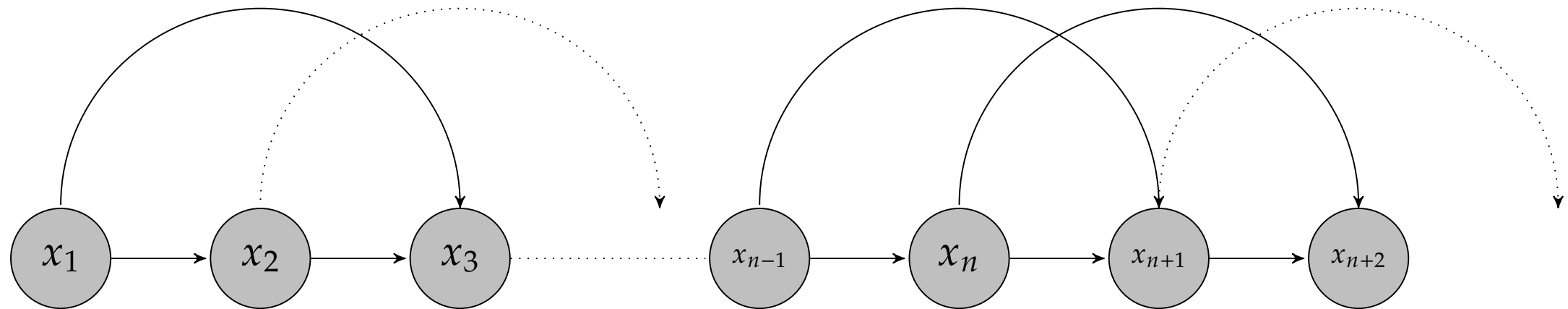


$$p(x_1, \dots, x_N) = p(x_1)p(x_2|x_1) \dots p(x_N|x_{N-1})$$

$$x_{n+1} \perp\!\!\!\perp x_{n-1} | x_n$$

time homogeneous $p(x_n|x_{n-1}) = p(x_2|x_1)$

Markov Models of order k



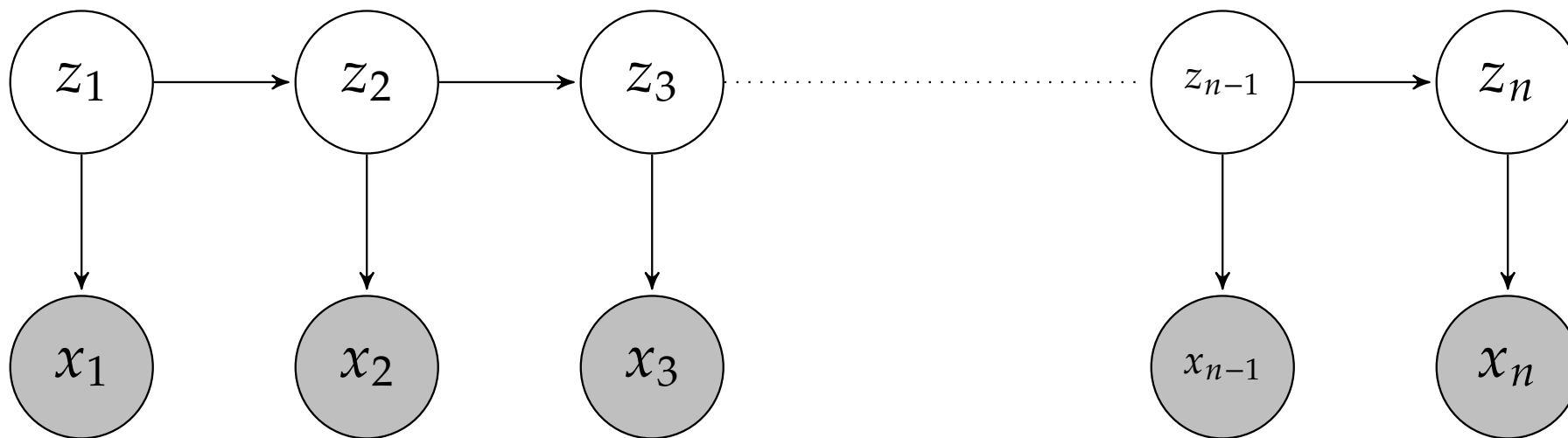
$$p(x_1, \dots, x_N) = p(x_1)p(x_2|x_1)p(x_3|x_2, x_1) \dots p(x_{n+1}|x_n, x_{n-1}) \dots$$

$$x_{n+2} \perp\!\!\!\perp x_{n-1} | x_n, x_{n+1}$$

x discrete: Markov Chain

x continuous: autoregressive model

State space models



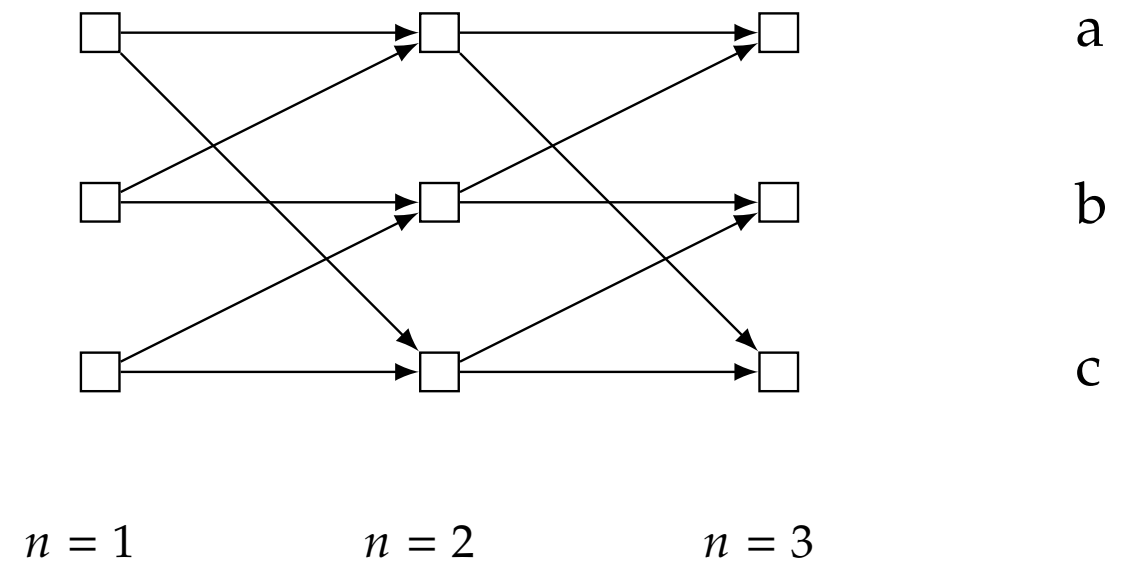
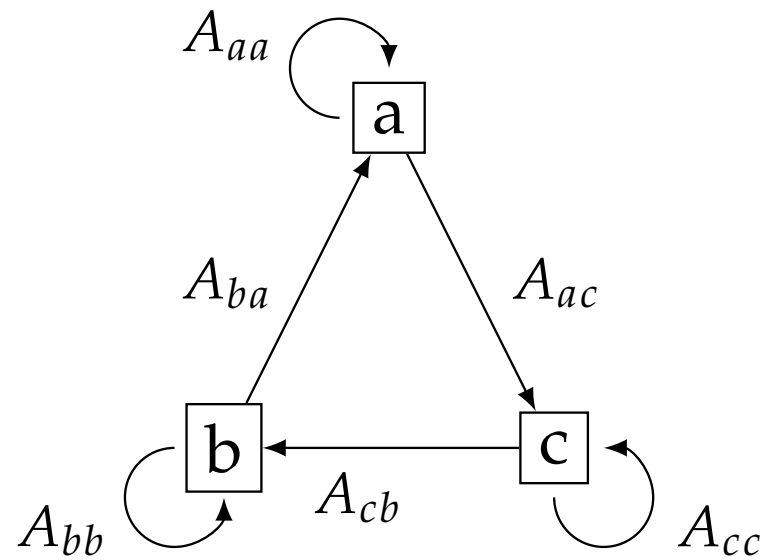
$$x_i \not\perp x_j | \underline{x}$$

- transition probabilities $p(z_n|z_{n-1})$
- initial distribution $p(z_1)$,
- emission probability $p(x_n|z_n)$.

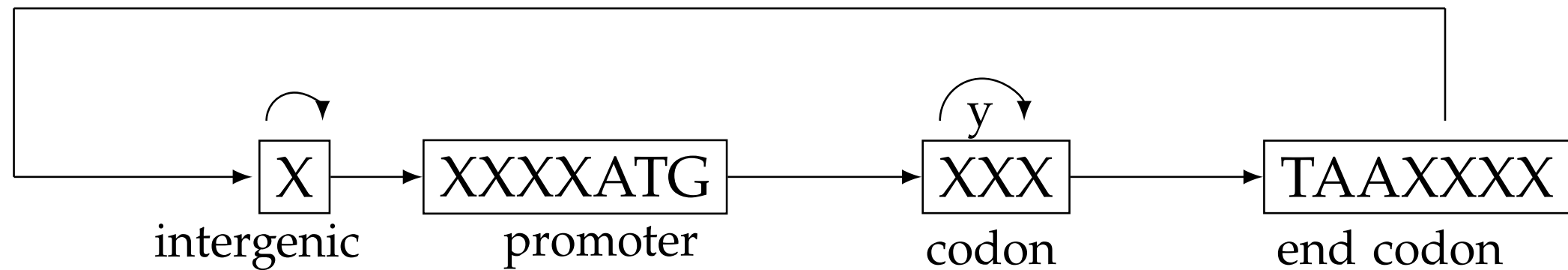
$$\theta = (A, \pi, \psi) \quad \underline{x}, \underline{z} = (x_1, \dots, x_n, z_1, \dots, z_n)$$

$$p(\underline{x}, \underline{z} | \theta) = p(z_1 | \pi) \left[\prod_{n=2}^N p(z_n | z_{n-1}, A) \right] \left[\prod_{n=1}^N p(x_n | z_n, \psi) \right]$$

State Space Models

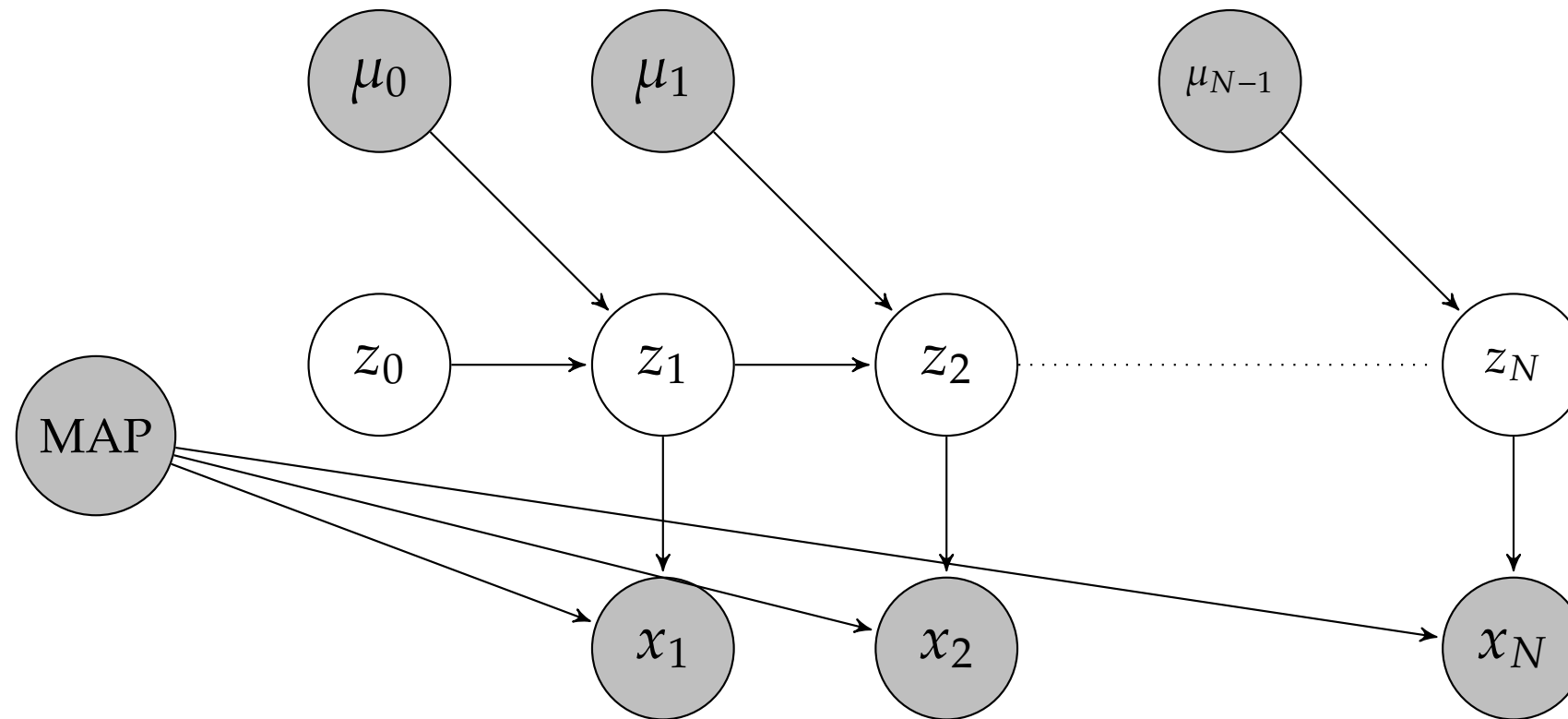


HMM for gene finding



with $x \in \{A, C, G, T\}$, $y \in \{[A, C, G, T]\}^3$.

HMM for robot localisation



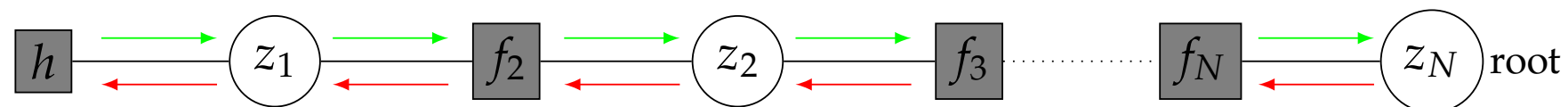
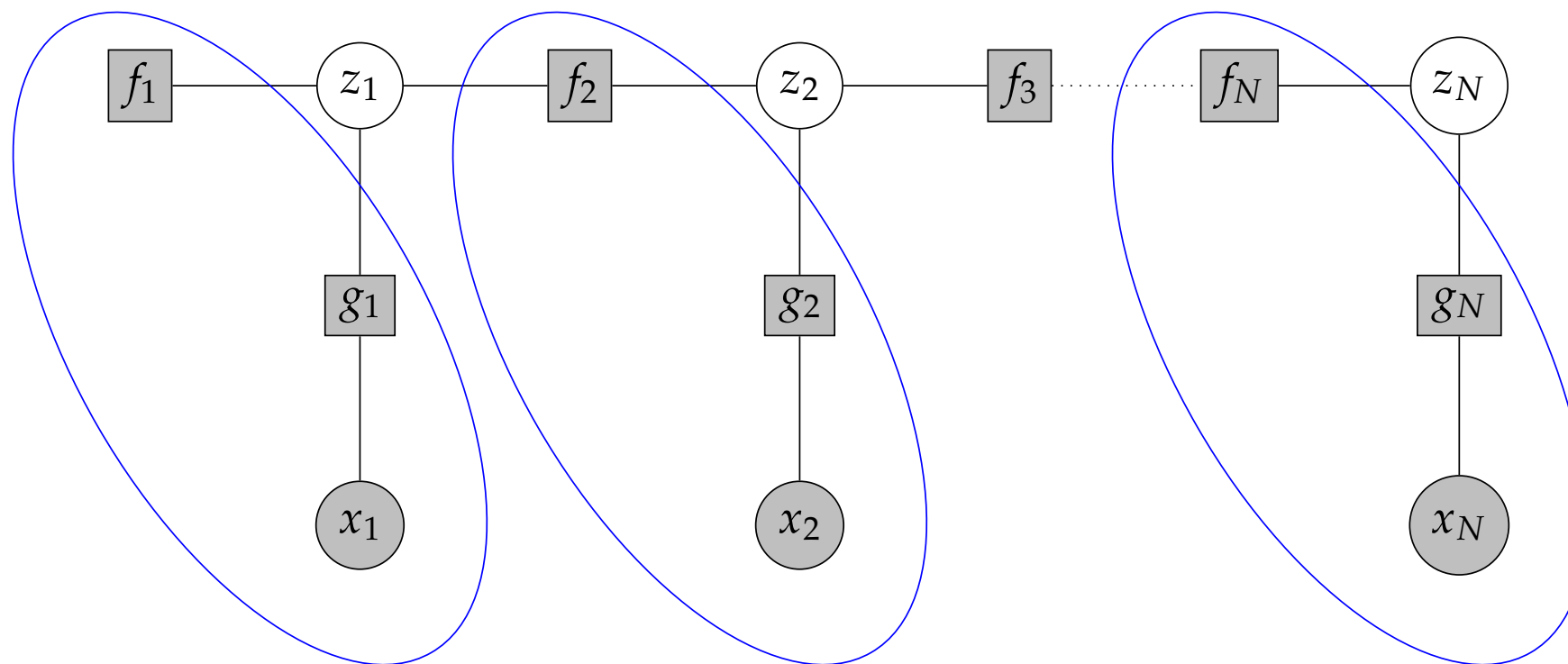
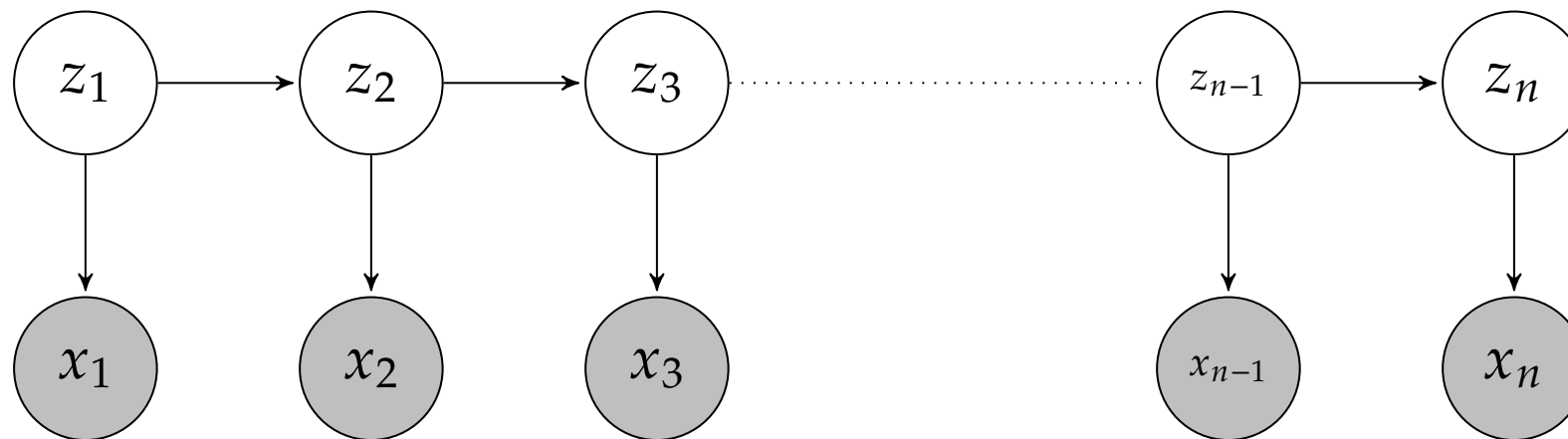
Inference in HMM

filtering

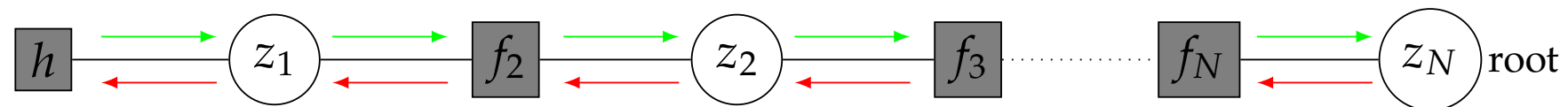
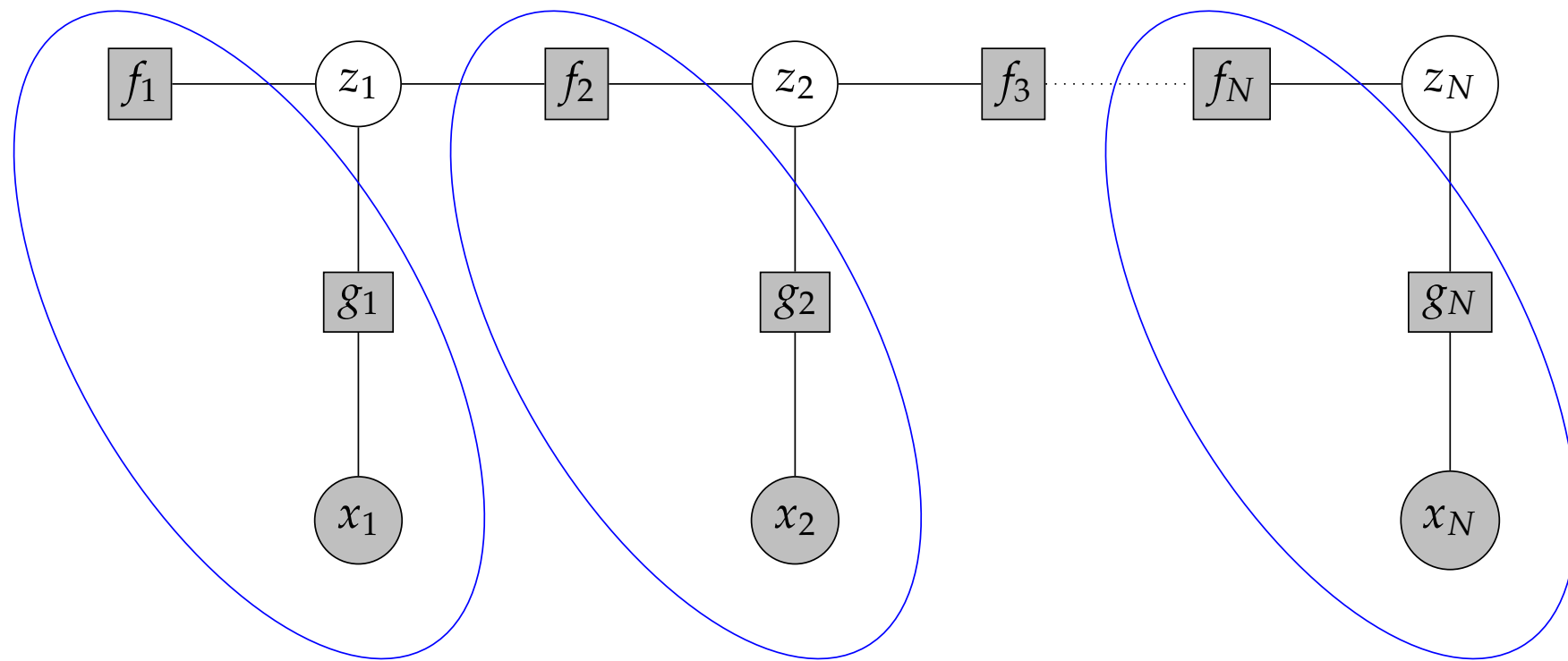
smoothing

most likely explanation

HMM to factor graphs



HMM to factor graphs

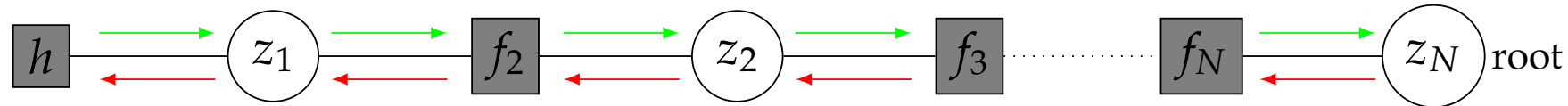


$$h(z_1) = p(z_1)p(x_1|z_1)$$

$$f_2(z_1, z_2) = p(z_2|z_1)p(x_2|z_2)$$

$$f_N(z_{N-1}, z_N) = p(z_N|z_{N-1})p(x_N|z_N)$$

Message Passing



- Forward messages (green arrows in the figure above):

$$\mu_{z_{n-1} \rightarrow f_n}(z_{n-1}) = \mu_{f_{n-1} \rightarrow z_{n-1}}(z_{n-1})$$

$$\mu_{f_n \rightarrow z_n}(z_n) = \sum_{z_{n-1}} f_n(z_{n-1}, z_n) \cdot \mu_{z_{n-1} \rightarrow f_n}(z_{n-1})$$

$$\alpha(z_n) := \mu_{f_n \rightarrow z_n}(z_n)$$

$$\alpha(z_n) = \sum_{z_{n-1}} f_n(z_{n-1}, z_n) \cdot \alpha(z_{n-1})$$

- Backward messages (red arrows in the figure above):

$$\mu_{f_{n+1} \rightarrow z_n}(z_n) = \sum_{z_{n+1}} f_{n+1}(z_n, z_{n+1}) \cdot \mu_{f_{n+2} \rightarrow z_{n+1}}(z_{n+1})$$

$$\beta(z_n) := \mu_{f_{n+1} \rightarrow z_n}(z_n)$$

$$\beta(z_n) = \sum_{z_{n+1}} f_{n+1}(z_n, z_{n+1}) \cdot \beta(z_{n+1})$$

Filtering and Smoothing

filtering $p(z_N, \underline{x}) = \alpha(z_N)$

smoothing $p(z_n, \underline{x}) = \alpha(z_n)\beta(z_n)$

$$p(z_n | \underline{x}) = \frac{p(z_n, \underline{x})}{\sum_{z_n} p(z_n, \underline{x})} = \frac{p(z_n, \underline{x})}{p(\underline{x})} = \frac{p(z_n, \underline{x})}{\sum_{z_n} \alpha(z_n)}$$

$$p(z_n, z_{n+1}, \underline{x}) = \alpha(z_n) f(z_n, z_{n+1}) \beta(z_{n+1})$$

Viterbi algorithm

most likely explanation

$$\hat{\mu}_{f_n \rightarrow z_n} = \max_{z_{n-1}} \{ \log f_n(z_n, z_{n-1}) + \hat{\mu}_{f_{n-1} \rightarrow z_{n-1}}(z_{n-1}) \}$$

$$\Phi(z_n) = \operatorname{argmax}_{z_{n-1}} \{ \log f_n(z_n, z_{n-1}) + \hat{\mu}_{f_{n-1} \rightarrow z_{n-1}}(z_{n-1}) \}$$

